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基於多尺度高解析度原型生成 YOLO 之細胞實例分割
YOLO-based Cell Instance Segmentation with Multi-Scale
High Resolution Prototype Generation

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摘要

自動化細胞實例分割是計算病理學中的關鍵步驟，為細胞結構分析、癌症分級和藥物開發等核心任務奠定了基礎。然而，標準的 YOLO 架構主要是針對自然影像進行優化，當應用於生物醫學顯微影像時，往往顯示出顯著的效能落差。具體而言，這些模型通常缺乏描繪微小細胞物件所需的空間細緻度，且難以有效區分密集聚落中的相連實例。為了解決這些挑戰，本論文提出了一個基於 YOLOv8-seg [1] 的框架，並引入了三項關鍵貢獻：高解析度原型網路以保留細緻的空間細節；原型導引係數生成機制以將偵測結果與結構先驗對齊；以及方向一致性損失以強制分離緊密相鄰個體。本方法在 Data Science Bowl 2018 和 PanNuke 資料集上進行了嚴格的評估。實驗結果顯示，相較於現有方法，所提出的方法達到了具競爭力的效能，並在聚合 Jaccard 指標和全景品質上展現了顯著的提升。這些研究發現表明，所提出的模型為多樣化生物情境中的細胞實例分割提供了一個穩健的解決方案。

關鍵字：深度學習、細胞實例分割、生物醫學影像分析、顯微影像分析



Abstract

Automated cell instance segmentation is a crucial step in computational pathology, forming the foundation for key tasks such as analyzing cell structures, cancer grading, and drug discovery. However, standard YOLO-based architectures are primarily optimized for natural images and reveal significant performance gaps when applied to biomedical microscopy. Specifically, these models often lack the spatial granularity required to delineate tiny cellular objects and exhibit limited capability in effectively disentangle touching instances in dense clusters. To address these challenges, this thesis proposes a novel framework based on YOLOv8-seg [1], introducing three key contributions: a High-Resolution Prototype Network (HRPN) to preserve fine-grained spatial details, a Proto-Guided Coefficient Generation mechanism to align detection with structural priors, and a Directional Consistency Loss to enforce separation between touching neighbors. The method is rigorously evaluated on the Data Science Bowl 2018 and PanNuke datasets. Experimental results indicate that the proposed method achieves competitive performance relative

to existing approaches, showing notable improvements in Aggregated Jaccard Index and Panoptic Quality. These findings suggest that the proposed model offers a robust solution for cell instance segmentation in diverse biological contexts.

Keywords: Deep Learning, Cell Instance Segmentation, Biomedical Image Analysis, Microscopy Image Analysis



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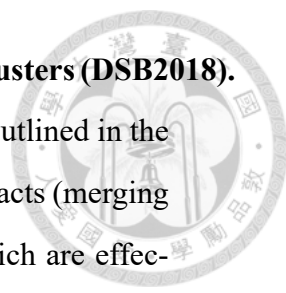




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Chapter 1 Introduction

Automated cell instance segmentation is a crucial step in computational pathology and biomedical image analysis. To clearly define the scope of this research, it is essential to distinguish instance segmentation from other fundamental computer vision tasks. While object detection localizes cells using bounding boxes and semantic segmentation classifies pixels into categories (e.g., cell vs. background) without distinguishing individual instances, instance segmentation integrates these objectives. It simultaneously detects distinct objects and delineates their precise boundaries. A visual comparison of these tasks is presented in Figure 1.1. This capability is particularly vital in biomedical contexts, where separating touching or overlapping cells is required for accurate quantification. It forms

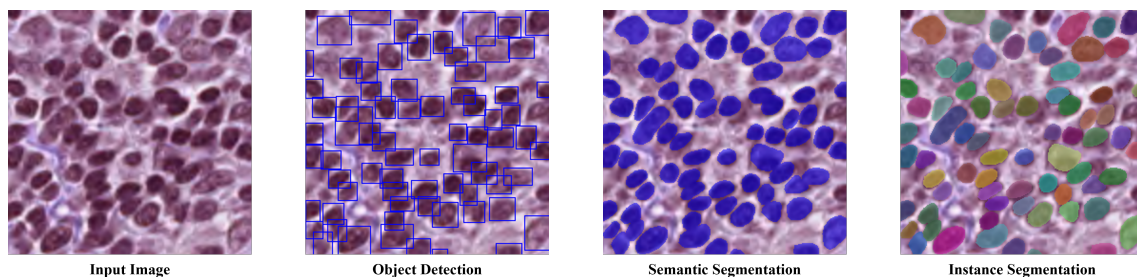


Figure 1.1: Illustration of different tasks in biomedical microscopy.

the foundation for key tasks such as analyzing cell structures, cancer grading, and drug discovery. Accurately outlining individual cell allows researchers to extract important visual features—such as size and shape, as demonstrated in Figure 1.2—which are essential for tracking disease progression and evaluating treatments. However, unlike objects in natu-

ral scenes, cells in microscopy images present unique challenges. They are often tightly packed, have unclear boundaries, and appear against messy and complex backgrounds. Consequently, developing robust algorithms capable of separating individual cells in such difficult environments remains a critical objective.

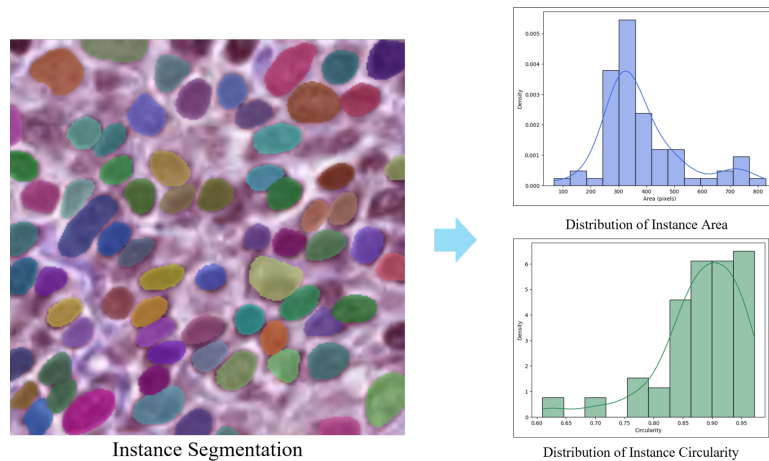


Figure 1.2: Quantitative morphological analysis derived from instance segmentation. The segmentation map (left) enables the extraction of precise metric, such as cell circularity and area distributions (right), facilitating statistical assessment of cell populations.

While deep learning approaches, particularly ”detect-then-segment” frameworks like Mask R-CNN [2], have established strong benchmarks in this domain, the field is increasingly exploring unified, one-stage architectures such as YOLO-based segmentation models [1, 3]. These models integrate localization and pixel-wise classification into a cohesive pipeline, offering significant advantages in architectural efficiency compared to multi-stage counterparts. However, standard YOLO models are primarily optimized for natural images (e.g., the COCO dataset [4]) where objects are typically large and distinct. Directly transposing this general-purpose architecture to biomedical microscopy reveals significant performance gaps. Specifically, the standard design often lacks the spatial granularity required to delineate the fine boundaries of tiny cellular objects and struggles to effectively disentangle touching instances in dense clusters.

1.1 Motivation



Three critical architectural bottlenecks are identified when transposing these general-purpose models to biomedical microscopy.

The first challenge is spatial precision. Standard YOLO-based architectures are inherently designed to prioritize high-level semantic abstraction to facilitate robust object categorization. However, a critical conflict arises when the feature maps resulting from aggressive downsampling are repurposed for pixel-wise segmentation. While low-resolution features are sufficient for bounding box regression, they result in an irreversible loss of spatial fidelity required for fine-grained mask generation. In cytology, where cells often appear as tiny objects occupying limited pixel resolution, this lack of granularity results in coarse masks that fail to capture the subtle morphological details essential for accurate clinical analysis.

The second limitation is the lack of structural consistency. In standard YOLO segmentation frameworks, the prediction of instance-specific parameters (coefficients) and the generation of global spatial templates (prototypes) occur independently. Consequently, the detection head predicts mask coefficients without being conditioned on the spatial priors encoded in the prototypes. This absence of alignment limits the model's ability to generate masks that are structurally compatible with the specific shape features of the detected cells, often leading to suboptimal reconstruction.

The third challenge is instance separation in crowded regions. Unlike natural scenes where objects are typically spatially distinct, biological tissues feature cells that are tightly clustered or directly adjacent to one another. Typical YOLO-based segmentation models

often lack a specific discriminative mechanism to distinguish the shared boundaries between these neighboring instances. Consequently, the model is prone to mask spillover between adjacent cells, failing to maintain distinct identities in biologically dense environments.

1.2 Contributions

To address the aforementioned challenges, this thesis proposes a novel cell instance segmentation framework based on the YOLOv8-seg architecture. The primary contributions of this work are summarized as follows:

1. **High-Resolution Prototype Network (HRPN):** A specialized prototype branch is constructed to generate high-fidelity mask templates. Unlike standard designs that rely on low-resolution feature maps, this network leverages Split-Attention Fusion blocks to aggregate multi-scale features while explicitly maintaining high-resolution representations. This design ensures the preservation of fine-grained spatial details, resulting in significantly sharper and more accurate cell boundaries.
2. **Prototype-Guided Coefficient Generation:** A Feature-wise Linear Modulation mechanism is introduced to bridge the informational gap between the coefficient head and the prototype branch. This module conditions the generation of mask coefficients on the spatial priors from the prototype network, ensuring that the predicted coefficients are structurally aligned with the global prototype features.
3. **Directional Consistency Loss (DCL):** An auxiliary loss function is designed to enforce directional alignment in the feature space. This loss explicitly optimizes the

cosine similarity to ensure that the coefficient vector of a specific instance aligns with its own prototype features, while simultaneously suppressing the similarity with touching neighbors towards negative values. This push-pull force effectively separates touching instances and reduces spillover errors.

4. **Comprehensive Evaluation:** Rigorous experiments are conducted on two diverse and challenging datasets: the Data Science Bowl 2018 [5] and the PanNuke dataset [6]. The proposed method demonstrates competitive performance compared to existing approaches, achieving favorable results in Aggregated Jaccard Index and Panoptic Quality, validating its robustness across both fluorescence microscopy and histopathology domains.

1.3 Thesis Organization

The remainder of this thesis is organized as follows:

- Chapter 2 reviews related work in cell segmentation, covering traditional image processing methods, non-detection-based deep learning approaches, and the evolution of detection-based frameworks, with a specific focus on YOLO architectures.
- Chapter 3 details the proposed methodology, including the HRPN, the FiLM interaction mechanism, and the DCL loss.
- Chapter 4 presents the experimental setup, a comprehensive analysis of quantitative and qualitative results, and ablation studies.
- Chapter 5 summarizes the contributions and discusses future research directions.



Chapter 2 Related Work

This chapter reviews the progression of methodologies for cell instance segmentation. It begins with traditional image processing techniques, focusing on non-learning-based approaches for extracting cellular structures from images. Subsequently, non-detection-based deep learning methods are examined, detailing how individual cell instances are localized and differentiated without relying on object detection frameworks. Finally, detection-based deep learning approaches are discussed, as illustrated in Figure 2.1. Special attention is given to the YOLOv8-seg architecture, where we analyze its mask generation mechanism and identify specific architectural limitations that hinder its performance in cellular instance segmentation.

2.1 Non-Deep Learning Approaches

Before the advent of deep learning, cell segmentation primarily relied on image processing techniques that exploited low-level features such as intensity, gradient, and shape geometry. These traditional approaches can be broadly categorized into region-based methods, such as the watershed transform, and boundary-based methods, such as active contour models.

The watershed transform treats image intensity as a topographic surface, where light

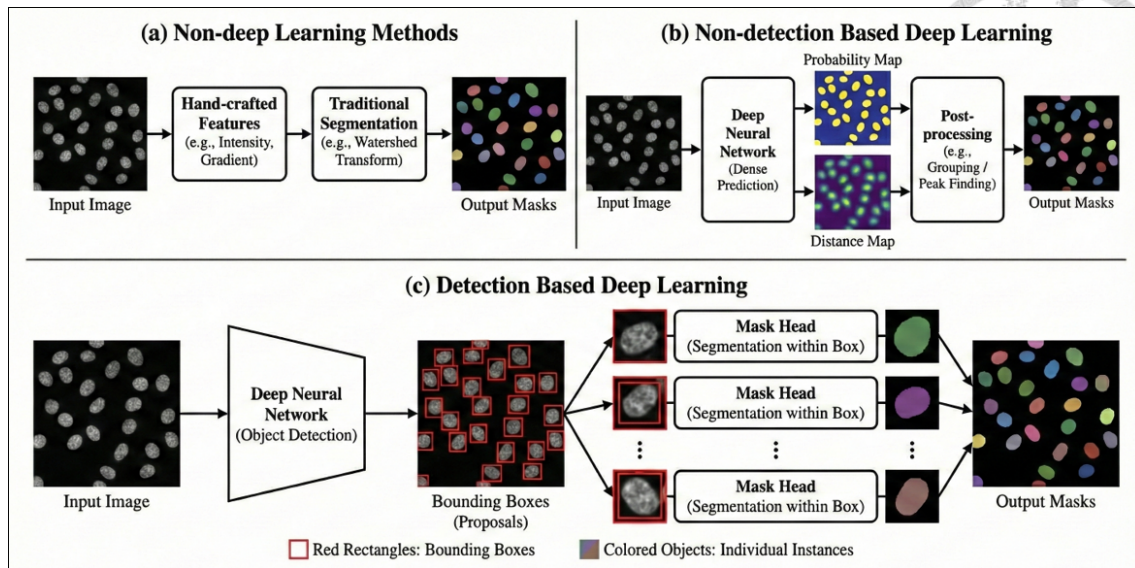


Figure 2.1: Schematic overview of cell instance segmentation methodologies. (a) Traditional non-deep learning methods rely on image processing techniques. (b) Non-detection based deep learning approaches formulate the task as dense pixel-wise prediction (e.g., distance maps). (c) Detection-based deep learning frameworks utilize bounding box localization prior to segmentation.

pixels represent high altitudes and dark pixels represent catchment basins. Boundaries are formed where waters from different basins meet. While effective for separating objects based on intensity topology, direct application often leads to over-segmentation due to image noise. To mitigate this, marker-controlled strategies were introduced. Malpica et al. [7] proposed using distinct marking strategies based on sample types, such as distance transforms for granular nuclei, to separate clustered cells. Similarly, Koyuncu et al. [8] developed "Smart Markers" by classifying pixels into groups based on visual characteristics to extract more robust markers for phase-contrast live cell segmentation.

Alternatively, Active Contour Models (Snakes) treat segmentation as an energy minimization problem, evolving a parametric curve to fit object boundaries based on internal smoothness and external image gradients. To address the sensitivity to initialization and local minima inherent in traditional snakes, Bamford and Lovell [9] introduced a dual active contour method using Viterbi search to find the global minimum energy path. Fur-

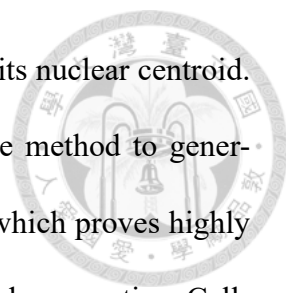
thermore, Hu et al. [10] proposed a polar coordinate-based active contour model to reduce computational complexity and improve boundary tracking. Despite these advancements, traditional methods remain limited by their reliance on hand-crafted features and parameter tuning, making them less adaptable to the high variability observed in large-scale biomedical datasets.

2.2 Non-Detection Based Deep Learning Approaches

Unlike detection-based methods that rely on bounding boxes, non-detection-based approaches typically formulate cell instance segmentation as a bottom-up, dense prediction task. These methods generally follow a two-stage logic: first, identifying foreground regions through binary segmentation mechanisms, and second, employing auxiliary pixel-wise predictions—such as distance maps, vector flows, or geometric descriptors—to disentangle individual instances within crowded clusters.

Contour-aware Methods. A critical challenge in separating touching histological structures is the lack of clear boundaries. To address this, DCAN [11] proposes a unified multi-task learning framework that simultaneously predicts probability maps for target objects and their contours. By explicitly learning contour features alongside the object mask, DCAN effectively separates clustered objects that would otherwise be merged in standard binary segmentation. This approach demonstrates the value of auxiliary geometric supervision in dense prediction tasks.

Distance and Flow-based Methods. A prominent strategy involves predicting pixel-wise geometric attributes to guide the separation of instances. In the context of histology, Hover-Net [12] proposed a unified network that simultaneously predicts a nuclear prob-



ability map and the horizontal and vertical distances of each pixel to its nuclear centroid. These distance maps encode rich instance information, allowing the method to generate energy landscapes for a marker-controlled watershed algorithm, which proves highly effective in splitting touching nuclei. Similarly, adopting a flow-based perspective, Cellpose [13] introduced a generalist approach derived from simulated diffusion. In addition to predicting a binary cell probability map, Cellpose predicts a spatial gradient flow field. During inference, pixels are grouped into instances by dynamically tracking these gradients towards their respective cell centers, a process uniquely capable of handling highly irregular and non-convex cell shapes.

Polygon-based Methods. Alternatively, polygon-based methods formulate the task as a geometric regression problem. This category is exemplified by StarDist [14], which represents nuclei as star-convex polygons by predicting radial distances from the object center. Unlike pixel-wise classification methods that typically require complex post-processing like watershed, StarDist directly generates instance shapes and relies on non-maximum suppression (NMS) to resolve overlapping candidates.

Graph-coloring Based Methods. Recently, a novel perspective inspired by the Four Color Theorem has been proposed to simplify instance differentiation. FCIS [15] reformulates the problem into a constrained semantic segmentation task. By assigning one of four distinct color labels to each cell such that no adjacent cells share the same color, instances can be separated without complex post-processing. Crucially, FCIS employs an asymptotic training strategy that explicitly prioritizes learning the binary foreground-background distinction before progressively learning the four-color categorical discrimination, ensuring training stability and effective instance separation.

2.3 Mask R-CNN Based Instance Segmentation



Detection-based approaches, widely referred to as top-down methods, operate by first localizing objects via bounding boxes and subsequently performing segmentation within these regions. While this paradigm theoretically distinguishes individual instances by assigning unique Regions of Interest (ROIs), it faces significant challenges in dense cytology scenarios. Specifically, when cells are closely aggregated, the resulting bounding boxes often highly overlap, causing the target ROI to be contaminated by features from adjacent instances. This target-interference ambiguity frequently leads to under-segmentation issues.

The most representative framework in this category is Mask R-CNN [2]. It extends the Faster R-CNN [16] object detector by adding a parallel branch for mask prediction. Specifically, the model generates region proposals and utilizes an RoIAlign mechanism to extract fixed-size feature maps for each candidate box, followed by a fully convolutional network (FCN) to predict the mask, as illustrated in Figure 2.2. However, within a

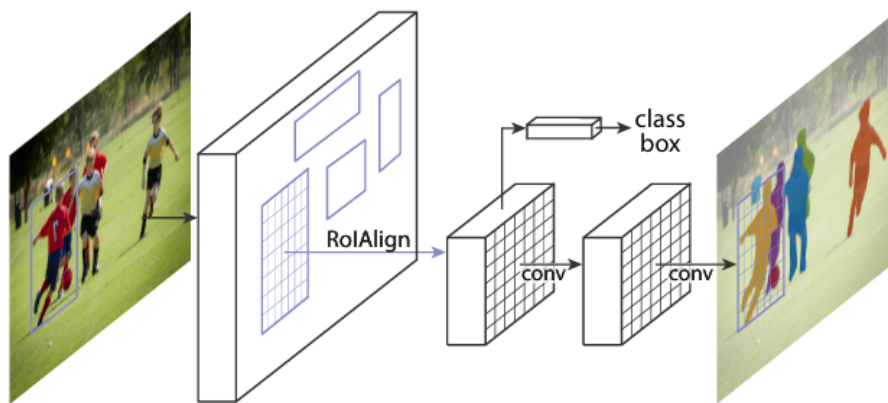
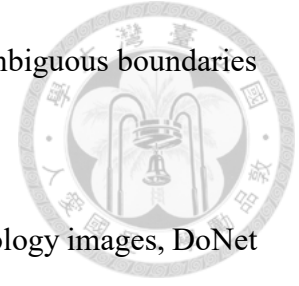


Figure 2.2: The Mask R-CNN framework for instance segmentation. The model extends Faster R-CNN by adding a parallel mask branch alongside the existing branch for classification and bounding box regression [2].

crowded ROI, the standard FCN branch lacks an explicit mechanism to suppress the fea-

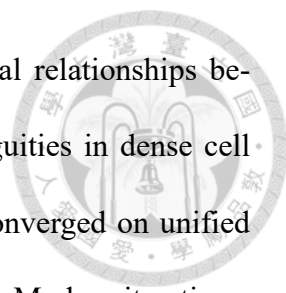
tures of interfering neighbors, often resulting in merged masks or ambiguous boundaries in touching regions.



To specifically address the challenge of overlapping cells in cytology images, DoNet [17] introduces a decompose-and-recombine strategy within the two-stage framework. Unlike standard Mask R-CNN, which treats occlusion implicitly, DoNet explicitly decouples cell clusters into target and interference features using a Dual-path Region Segmentation Module (DRM). It then employs a Semantic Consistency-guided Recombination Module (CRM) to analyze the intersection and complement regions between these decoupled components, ensuring semantic consistency and accurate boundary delineation for overlapping instances.

2.4 YOLO-based Instance Segmentation

While the YOLO architecture [18, 19] was originally established as a prominent framework for object detection, the research community has increasingly explored its potential for instance segmentation. This evolution has been particularly influential in the biomedical domain, where the need for efficient analysis of large-scale microscopy data is paramount. Early adaptations typically followed a cascaded "detect-then-segment" paradigm. For instance, frameworks such as YUSEG [20] and YOLO-LOGO [21] utilize YOLO models strictly as object localizers to generate regions of interest (ROIs). These ROIs are cropped and subsequently fed into separate, independent networks (e.g., U-Net or Transformers) for pixel-wise segmentation. Although this strategy allows the segmentation module to focus exclusively on the foreground, it suffers from a critical limitation regarding global context. Because the segmentation network operates solely on cropped



patches, it loses access to the broader image context and the spatial relationships between neighboring instances, which are crucial for resolving ambiguities in dense cell clusters. To address these limitations, recent advancements have converged on unified frameworks where detection and segmentation are optimized jointly. Modern iterations, specifically YOLOv5-seg [3] and YOLOv8-seg [1], have integrated segmentation heads directly into the one-stage architecture. Notably, these segmentation heads are inspired by the YOLACT [22] prototype generation mechanism, enabling the models to predict instance masks efficiently within a single cohesive network without the need for intermediate cropping or separate inference steps.

2.4.1 YOLOv8-seg Architecture

Overall Network Architecture. The YOLOv8-seg architecture, as illustrated in Figure 2.3, adopts a unified, anchor-free design that streamlines feature extraction and mask prediction into a cohesive pipeline. The framework comprises four key components: the Backbone, Neck, Head, and Post-processing modules. For feature extraction, the Backbone employs the C2f module as its primary building block. The C2f module is designed to enhance gradient flow and computational efficiency by combining the strengths of the Cross-Stage Partial [23] strategy with the ELAN architecture [24], allowing for richer feature embedding without excessive parameters. To ensure the network captures features at various granularities, the Spatial Pyramid Pooling Fast (SPPF) module is applied at the end of the backbone. This module efficiently fuses multi-scale local and global context by pooling features at sequential kernel sizes, improving the model's robustness to object scale variations. The Neck utilizes a PANet structure [25] to fuse these multi-scale features, which are then fed into a decoupled Head. Unlike pure detection models, the

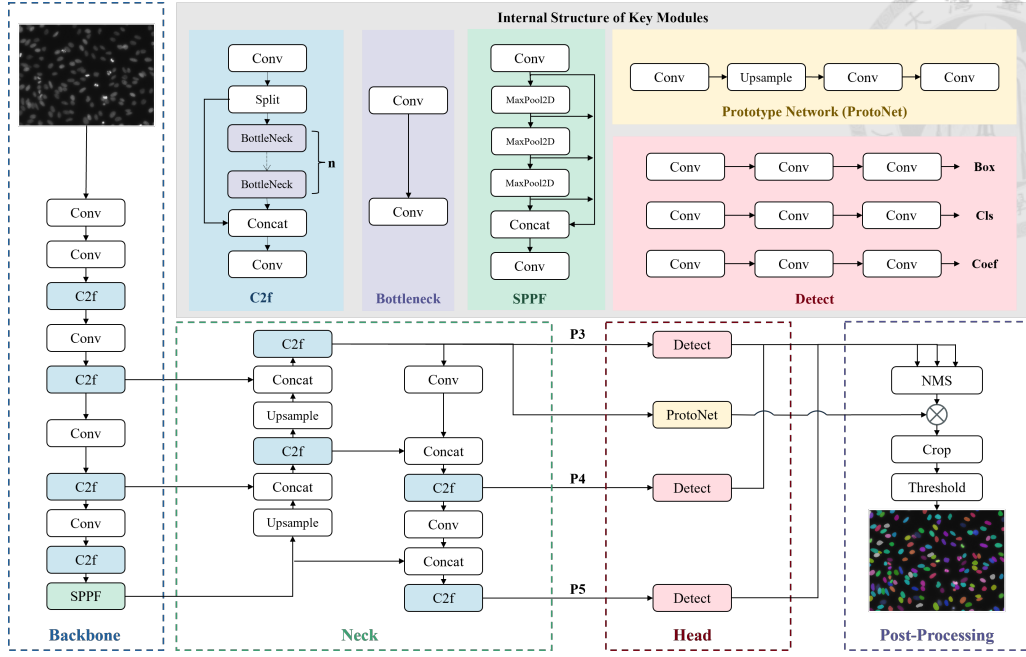
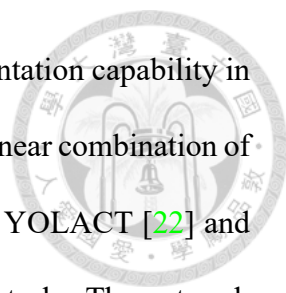


Figure 2.3: Architecture of YOLOv8-seg.

YOLOv8-seg head branches into three distinct tasks: object classification, bounding box regression, and mask coefficient prediction. Simultaneously, a separate ProtoNet branch operates in parallel to generate a set of global prototype masks. This architectural design necessitates a composite loss function to optimize all tasks jointly:

$$Loss = \lambda_{box}(L_{CIoU} + L_{DFL}) + \lambda_{cls}L_{cls} + \lambda_{seg}L_{seg} \quad (2.1)$$

Specifically, the box regression is optimized using CIoU [26] and Distribution Focal Loss (DFL) [27], while the segmentation task (L_{seg}) minimizes the error between the ground truth and the linear combination of the predicted coefficients and prototypes. DFL reformulates the regression as a discretized classification task over an interval $[0, \text{reg_max}]$, allowing the network to capture the uncertainty of predicted coordinates—a feature particularly beneficial for delineating cells with ambiguous boundaries. Finally, during inference, post-processing steps including Non-Maximum Suppression (NMS) and matrix operations are applied to assemble the final instance masks.



Mask Generation Mechanism. The core of the instance segmentation capability in YOLOv8-seg lies in its mask generation mechanism, which acts as a linear combination of prototypes and coefficients. This mechanism, originally proposed in YOLACT [22] and illustrated in Figure 2.4, fundamentally decouples the segmentation task. The network employs a ProtoNet branch to generate a set of k distributed prototype masks, denoted as $P \in \mathbb{R}^{H \times W \times k}$, which capture the global spatial patterns of the image. Simultaneously, the detection head predicts a set of instance-specific mask coefficients $c_i \in \mathbb{R}^k$ for each detected object i . The final raw mask M_{raw} is synthesized by computing a linear combination of the prototypes weighted by the predicted coefficients, followed by a sigmoid activation function to map values to the $(0, 1)$ range. This process can be mathematically formulated as:

$$M_{\text{raw}} = \sigma\left(\sum_{z=1}^k c_z P_z\right) \quad (2.2)$$

where σ denotes the sigmoid function, c_z is the z -th coefficient and P_z is the z -th prototype map. It is important to note that M_{raw} represents a full-image mask. To obtain the final instance mask M_{final} , the raw mask is cropped using the predicted bounding box coordinates and then thresholded (typically at 0.5) to binarize the pixel values. This distinct "assembly" approach allows the model to produce high-resolution masks without the computational burden of pixel-wise classification for every instance in the feature map.

Limitations of the Baseline YOLOv8-seg Framework. Despite the efficiency and accuracy of YOLOv8-seg in general object segmentation, directly applying it to cell instance segmentation reveals several limitations inherent to its general-purpose design. These limitations can be summarized as follows:

- **Insufficient Spatial Granularity in Prototypes.** The prototype generation branch

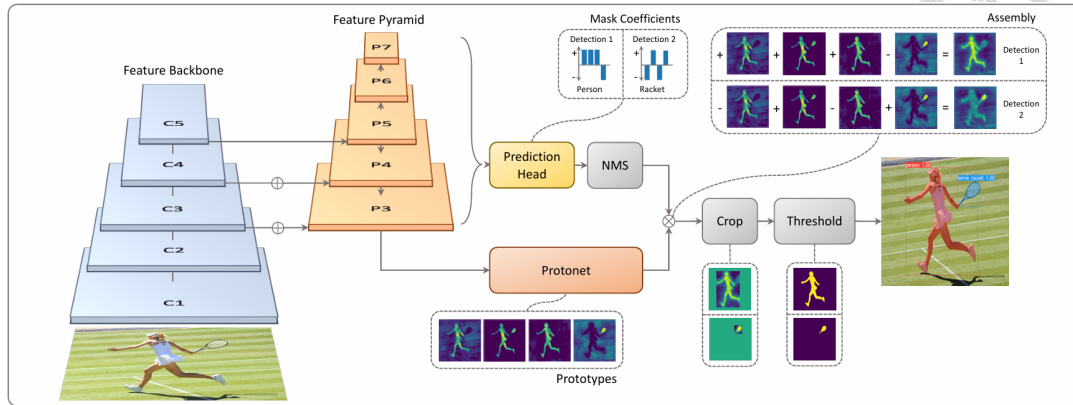


Figure 2.4: **Conceptual illustration of the mask generation mechanism originally proposed in YOLACT [22].** The YOLOv8-seg model adapts this parallel branch design (ProtoNet and Coefficients) to enable instance segmentation.

in the standard YOLOv8-seg architecture typically operates on feature maps derived from deep layers of the Feature Pyramid Network (FPN). While these features are semantically rich, they are the product of aggressive downsampling, which filters out high-frequency spatial details. Crucially, simply upsampling these features to a larger resolution cannot recover the lost spatial granularity. In the context of cytology, where cells are densely packed and require precise boundary delineation, this lack of inherent spatial fidelity results in mask prototypes that exhibit coarse or jagged boundaries. Although architectures designed to maintain high-resolution representations throughout the network—such as HRNet [28]—have proven effective in other dense prediction tasks, such mechanisms have not yet been effectively integrated into the prototype branches of one-stage instance segmentation frameworks.

- **Decoupled Prototype and Coefficient Generation.** In the original architecture, the generation of mask coefficients (by the detection head) and prototype masks (by the ProtoNet) occurs in isolated branches. The detection head predicts coefficients without any direct awareness of the spatial patterns encoded in the prototypes. This lack of interaction limits the model’s ability to adaptively weigh prototypes to cap-

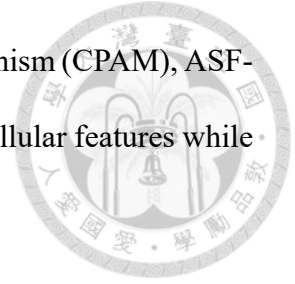
ture the subtle morphological variations of individual cell, resulting in suboptimal boundary reconstruction.

- **Absence of Directional Consistency Constraints.** Standard pixel-wise loss functions (e.g., Binary Cross Entropy) used in YOLOv8-seg primarily drive pixel-level classification, potentially underemphasizing the angular alignment between instance-specific coefficients and spatial prototypes. Lacking this explicit directional constraint, the network struggles to distinguish the directional signatures of adjacent instances. In crowded cytology images, this angular ambiguity often results in mask spillover, where the failure to decouple instance features causes predicted boundaries to encroach upon neighboring cells.

2.4.2 YOLO-based Models for Cell Instance Segmentation

Applying YOLO-based architectures to biomedical imagery presents unique challenges, primarily due to the small size, dense packing, and weak boundaries characterizing cellular structures. Consequently, adaptations in this domain typically focus on enhancing feature extraction capabilities and incorporating attention mechanisms to suppress background noise while highlighting fine-grained cellular details. YOLO-TL [29] improves upon the YOLOv8-seg architecture to handle low-quality medical images. It integrates space-to-depth (SPD) convolution and parameter-free attention mechanisms (SimAM) [30] into the backbone. By replacing standard strided convolutions, this design prevents the loss of fine-grained features during downsampling, enabling more accurate segmentation of tiny nuclei. Similarly, ASF-YOLO [31] builds on the YOLOv5-seg to address the difficulties of segmenting small and dense cells. It introduces a Scale Sequence Feature Fusion (SSFF) module and a Triple Feature Encoder (TFE) to enhance multiscale feature

extraction. By incorporating a Channel and Position Attention Mechanism (CPAM), ASF-YOLO effectively directs the network's focus towards informative cellular features while maintaining the efficiency of the unified YOLO architecture.





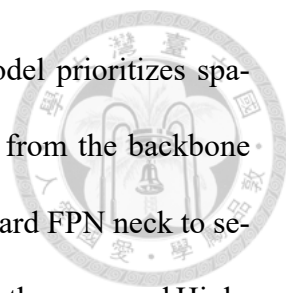
Chapter 3 Method

This chapter presents the proposed cell instance segmentation framework, which improves upon the YOLOv8-seg baseline. To address the challenges of (1) fine-grained cell delineation, (2) cross-branch feature alignment, and (3) instance separation in dense clusters, this study introduces three core contributions: a High-Resolution Prototype Network (HRPN), a Proto-Guided Coefficient Generation mechanism, and a Directional Consistency Auxiliary Loss. Section 3.1 provides a holistic overview of the system architecture. Section 3.2 details the design of the HRPN, followed by the interaction mechanism for coefficient generation in Section 3.3. Finally, Section 3.4 describes the auxiliary loss designed for robust optimization.

3.1 Overview

The overall architecture of the proposed cell instance segmentation model is illustrated in Figure 3.1. Built upon the YOLOv8-seg baseline, the proposed framework retains the original backbone and PAFPN neck for robust feature extraction but fundamentally reconstructs the mask generation pipeline to address the challenges of fine-grained cell delineation. The proposed method introduces three key architectural enhancements:

High-Fidelity Feature Aggregation. Unlike the standard architecture that relies



on a lightweight, low-resolution prototype branch, the proposed model prioritizes spatial fidelity from the bottom up. The multi-scale features extracted from the backbone are bifurcated into two pathways. The first pathway follows the standard FPN neck to semantic abstraction for object detection. The second pathway feeds into the proposed High-Resolution Prototype Network (HRPN). This specialized sub-network explicitly preserves the fine-grained spatial details from the shallow layers (C_2) while integrating semantic context through cascaded Split-Attention Fusion blocks. This ensures that the resulting prototype masks maintain high-fidelity boundaries essential for separating dense cell instances.

Prototype-Guided Coefficient Generation. To resolve the informational disconnect between the detection and segmentation tasks, a bridge is introduced between the two parallel pathways. Instead of generating mask coefficients in isolation, the detection head is conditioned on the structural features from the HRPN. This mechanism utilizes Feature-wise Linear Modulation (FiLM) [32] to inject prototype-aware guidance into the coefficient branch, ensuring that the generated coefficients are structurally compatible with the high-resolution prototypes.

Directional Consistency Optimization. Finally, to enforce strict feature alignment during training, a Directional Consistency Loss (DCL) is introduced. Beyond standard YOLOv8-seg multitask objectives, this auxiliary loss explicitly prioritizes angular alignment over feature magnitude between the generated coefficients and prototypes. By strictly enforcing this directional consistency, the model can effectively distinguish the unique signatures of adjacent instances, ensuring robust segmentation even in densely packed environments.

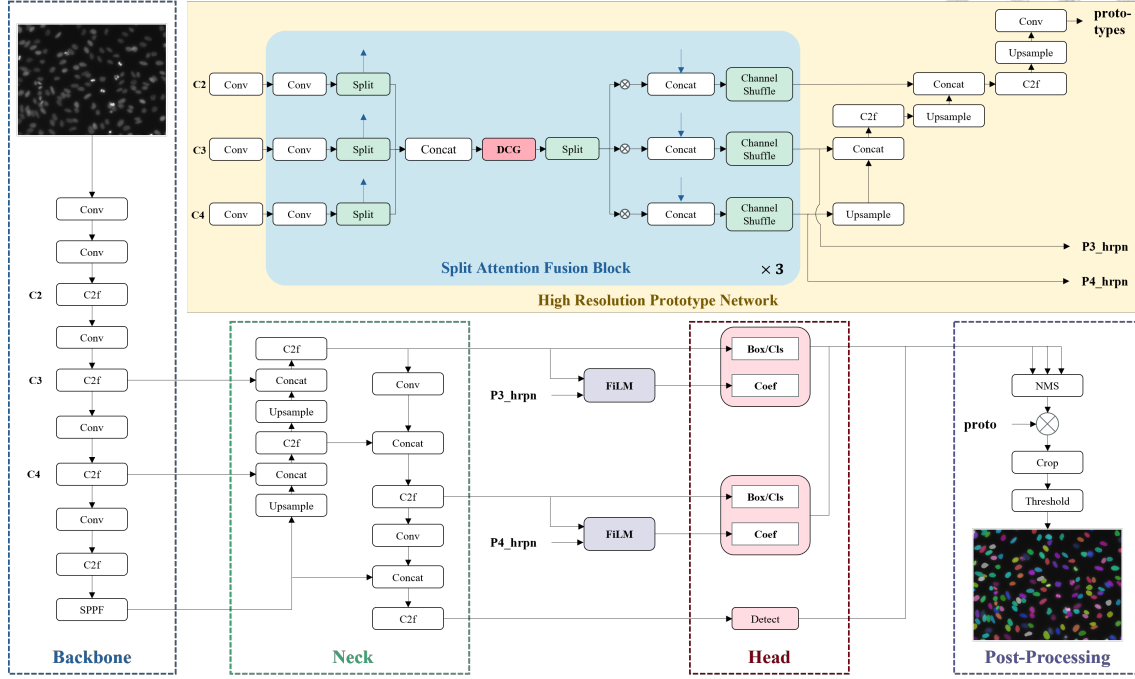


Figure 3.1: Architecture of Proposed Model

3.2 High-Resolution Prototype Network

This section presents the High-Resolution Prototype Network (HRPN), a specialized sub-network designed to generate high-fidelity prototype masks and features for the cell instance segmentation head. Unlike standard Feature Pyramid Networks (FPN) that often sacrifice spatial details for semantic abstraction, the proposed HRPN is constructed to maintain high-resolution representations throughout the feature aggregation process. This design philosophy draws inspiration from HRNet [28], which demonstrated the efficacy of maintaining parallel multi-resolution streams for position-sensitive tasks. However, this architecture is adapted specifically as an efficient prototype generator rather than a heavy backbone. Taking multi-scale features $\{C_2, C_3, C_4\}$ from the backbone as input, the network focuses on preserving fine-grained information at the C_2 scale while integrating deep semantic context. The architectural details and the proposed interaction mechanisms are described in the following subsections.

3.2.1 Network Architecture



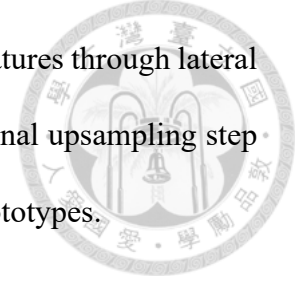
The HRPN operates on the multi-scale features extracted from the backbone, specifically denoted as $\{C_2, C_3, C_4\}$. In this configuration, the focus is placed exclusively on these high-to-medium resolution branches to prioritize spatial precision for mask generation, omitting the coarsest C_5 feature to reduce computational redundancy. The overall pipeline consists of three distinct stages: feature alignment, interactive encoding, and progressive reconstruction.

Feature Alignment. Initially, the input features $\{C_2, C_3, C_4\}$ are individually processed by convolution layers. This step aligns the disparate channel dimensions of the backbone outputs into a unified feature space, preparing them for the subsequent interaction steps.

Interactive Encoding (Stacked SAF). The core of the HRPN is the encoder, which is composed of N cascaded Split-Attention Fusion (SAF) Blocks (detailed in Section 3.2.2). These blocks are designed to facilitate robust feature interaction across different spatial scales while explicitly preserving the fine-grained local details of each feature layer. In each interaction step, the SAF block allows the high-resolution features to absorb semantic context from other scales through a selective attention mechanism, while simultaneously preserving local spatial details essential for boundary delineation. This hierarchical design ensures that the network captures both boundary information and semantic consistency before the final reconstruction stage.

Progressive Reconstruction. To generate the final prototype features, the multi-scale outputs from the final SAF Block are fused using a U-Net-like architecture [33]. The decoder follows a top-down pathway, where coarser semantic features are progres-

sively upsampled and merged with corresponding high-resolution features through lateral connections. The final aggregated feature map undergoes an additional upsampling step followed by a convolution layer to yield the final high-resolution prototypes.



3.2.2 Split-Attention Fusion Block

As illustrated in Figure 3.1, the Split-Attention Fusion (SAF) block serves as the fundamental building unit across all feature scales to model cross-scale dependencies while explicitly preserving the original feature information. Let $X \in \mathbb{R}^{C \times H \times W}$ denote the input feature map for any specific scale after the initial convolution refinement. To strictly preserve the fine-grained spatial details essential for delineating weak cellular boundaries, the design philosophy prioritizes feature retention over mere abstraction. Consequently, a channel splitting strategy is adopted: half of the feature channels are isolated as a “retention branch” to maintain original spatial fidelity, while the remaining channels undergo contextual transformation. While this structural topology draws inspiration from LiteHRNet [34], the internal processing is fundamentally distinct. The original LiteHRNet relies on adaptive average pooling for the transformation stage, which collapses spatial dimensions. However, this global aggregation is identified as too coarse for cytology, as it inevitably eliminates crucial local details. Therefore, the interaction mechanism is reformulated: instead of pooling, bilinear interpolation is employed to preserve the spatial layout, coupled with a Dilated Context Gating (DCG) mechanism. This ensures that while the split-branch structure is leveraged for feature preservation, the interaction branch generates a pixel-wise attention map, enabling dense, location-specific feature modulation. The internal processing workflow of the SAF block proceeds as follows. First, the input feature X undergoes a 3×3 convolution specifically for local feature extraction. Subse-

quently, the mechanism operates through three phases:

Channel Splitting. To protect the integrity of the input representation, X is split along the channel dimension into two parallel branches: the retention branch X_{retain} and the interaction branch $X_{interact}$. Specifically, the dimensions are defined as:

$$X_{retain}, X_{interact} \in \mathbb{R}^{\frac{C}{2} \times H \times W}$$

The retention branch X_{retain} is kept unchanged during the interaction phase. This separation ensures that the scale-specific details—whether fine-grained information in shallower layers or semantic patterns in deeper layers—are preserved directly in the output.

Context-Aware Interaction. The interaction branch $X_{interact}$ is responsible for capturing contextual information. To effectively capture global semantic context by expanding the effective receptive field, the feature is first downsampled via bilinear interpolation to a unified low-resolution scale (e.g., C4) and concatenated with aligned features from other scales. This aggregated representation is fed into the Dilated Context Gating (DCG) module (detailed in Section 3.2.3) to generate a spatial attention map $W \in \mathbb{R}^{\frac{C}{2} \times H \times W}$. The generated weights are partitioned back into their corresponding branches and subsequently upsampled to match the original spatial dimensions of the input. Finally, the feature is modulated by the resolution-restored weights:

$$\tilde{X}_{interact} = X_{interact} \odot W$$

where $\odot$ denotes element-wise multiplication. Here, the attention map W encapsulates multi-scale context from both the current layer and other layers, allowing the network to selectively reweight the features based on the aggregated context.



Aggregation and Shuffle. Finally, concatenating the retention branch X_{retain} and the refined interaction branch $\tilde{X}_{interact}$ forms the output. To prevent information segregation between the two branches in subsequent layers, a Channel Shuffle operation [35] is applied. This mechanism conceptually interlaces the channels from the retention and interaction groups, ensuring that feature channels are fully mixed before entering the next processing stage:

$$Y = \text{ChannelShuffle}(\text{Concat}(X_{retain}, \tilde{X}_{interact}))$$

The output Y thus synergizes the preserved local details from the retention branch with the globally contextualized semantics from the interaction branch.

3.2.3 Dilated Context Gating

The Dilated Context Gating (DCG) module serves as the core engine for cross-scale information integration. It is designed to process the unified low-resolution tensor X_{joint} obtained from the previous concatenation step and generate a refined weight map W_{joint} encapsulating global structural guidance. The architecture of DCG is illustrated in Figure 3.2. Given the input $X_{joint} \in \mathbb{R}^{C_{in} \times H \times W}$, the processing pipeline consists of three key stages: channel fusion, multi-scale context aggregation, and attention generation.

Channel Fusion and Local Encoding. Since X_{joint} is formed by concatenation, the channel information is initially segregated. A point-wise convolution (1×1) is first employed to blend the cross-scale channels. This is followed by a standard depthwise convolution (3×3) to encode basic local structural patterns F_{base} :

$$F_{base} = \text{DWConv}_{3 \times 3}(\text{Conv}_{1 \times 1}(X_{joint}))$$

This combination ensures that channel interaction and spatial encoding are handled sequentially.

Multi-Scale Context Aggregation. To capture contextual dependencies at varying ranges, parallel dilated depthwise convolutions are utilized. The feature F_{base} branches into three paths: an identity path (local), and two dilated paths with rates $d = 2$ and $d = 3$. The outputs are aggregated via element-wise summation:

$$F_{multi} = F_{base} + \text{DWConv}_{d=2}(F_{base}) + \text{DWConv}_{d=3}(F_{base})$$

This summation strategy fuses information from multiple receptive fields into a single representation.

Attention Generation. Since the aggregation is performed via summation, a final transformation is necessary to refine the fused features. A 1×1 convolution is applied to adjust the channel dimensions and integrate the summed signals, followed by a Sigmoid activation to generate the attention weights:

$$W_{joint} = \sigma(\text{Conv}_{1 \times 1}(F_{multi}))$$

The resulting W_{joint} serves as the global attention map, ready to be partitioned and distributed back to the respective feature branches. This mechanism ensures that the final spatial modulation is driven by a comprehensive synthesis of local details and multi-scale global context.

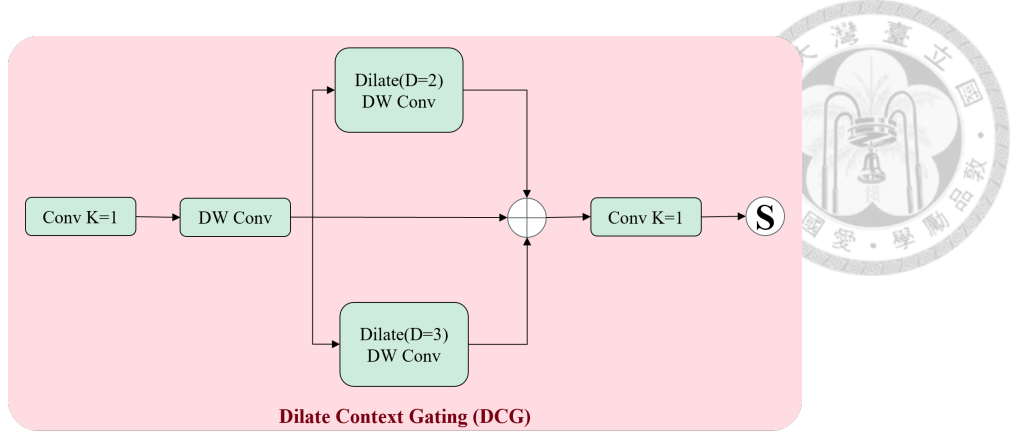


Figure 3.2: Architecture of Dilated Context Gating (DCG).

3.3 Prototype-Guided Coefficient Generation

To bridge the informational gap between the decoupled prototype and coefficient branches, this study introduces a modulation mechanism that conditions the coefficient generation on the structural priors from the HRPN. Unlike standard architectures where coefficients are generated independently, the proposed approach leverages Feature-wise Linear Modulation (FiLM) [32] to inject prototype guidance into the mask coefficient branch.

Mechanism and Data Flow. Specifically, the spatially rich features from the final SAF block of the HRPN are utilized as the condition, while the feature maps from the FPN neck serve as the target. Let $F_{neck} \in \mathbb{R}^{C \times H \times W}$ be the input feature for the coefficient head and F_{proto} be the corresponding guidance feature. The modulation is performed via an affine transformation:

$$\text{FiLM}(F_{neck}|F_{proto}) = (1 + \gamma(F_{proto})) \odot F_{neck} + \beta(F_{proto}) \quad (3.1)$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are the scale and shift parameters predicted by a lightweight generator network (a 1×1 convolution), and $\odot$ denotes channel-wise multiplication. This operation

allows the network to selectively emphasize or suppress feature channels in the detection head based on the semantic structure encoded in the prototype branch.

Scale Selection and Optimization. It is worth noting that this guidance is applied selectively to the P3 and P4 scales. Since cell instances are typically small objects requiring fine-grained spatial resolution, the coarser P5 scale (1/32 resolution) provides negligible structural benefit for mask generation and is thus excluded from this modulation to maintain efficiency. Finally, to preserve the pre-trained feature distribution of the detector during the initial training phase, a Zero Initialization strategy is employed. The final convolution layers predicting γ and β are initialized with zeros and biases, respectively. This ensures the module starts as an identity mapping:

$$\text{FiLM}_{init}(F_{neck}) = (1 + 0) \odot F_{neck} + 0 = F_{neck}$$

By initializing the modulation parameters to null, the network retains the original feature distribution at the start of training. This allows the model to gradually inject the prototype guidance without abruptly altering the feature distribution fed into the coefficient head.

3.4 Directional Consistency Loss

In the proposed framework, the final instance mask is reconstructed via a linear combination of prototype features and instance-specific coefficients. For any specific instance i and spatial location (x, y) , this operation is mathematically defined as the dot product between the coefficient vector $\mathbf{c}_i$ and the prototype vector $\mathbf{p}_{x,y}$:

$$m_{i,x,y} = \mathbf{c}_i \cdot \mathbf{p}_{x,y} = \|\mathbf{c}_i\| \|\mathbf{p}_{x,y}\| \cos \theta$$

During inference, since the mask is generated by thresholding the sigmoid probability at 0.5 (implying a positive logit), the segmentation decision relies heavily on the sign of this product—determined solely by $\cos \theta$.

However, the primary Binary Cross-Entropy (BCE) loss fundamentally optimizes the magnitude of the logits. This creates a potential geometric ambiguity: the network can minimize the loss by simply inflating the scale components ($\|\mathbf{c}\|$ and $\|\mathbf{p}\|$), even when the directional alignment is weak (i.e., $\theta \approx 90^\circ$, implying vanishing directional similarity). Such "high-scale but misaligned" representations are unstable, as minor perturbations can flip the sign of the dot product, resulting in fragmented or noisy masks. To explicitly decouple directional alignment from magnitude scaling, an auxiliary loss is proposed.

3.4.1 Directional Consistency Constraint

This objective is formulated using a margin-based Cosine Hinge Loss. The loss is computed strictly within the bounding box of the target instance i , focusing solely on distinguishing the instance from its interfering neighbors. To strictly isolate the directional alignment from magnitude variations, the objective is defined using Cosine Similarity between the coefficient vector $\mathbf{c}_i$ and the prototype vector $\mathbf{p}_{x,y}$:

$$S_{i,x,y} = \frac{\mathbf{c}_i \cdot \mathbf{p}_{x,y}}{\|\mathbf{c}_i\|_2 \|\mathbf{p}_{x,y}\|_2}$$

where $S_{i,x,y} \in [-1, 1]$. Two active regions are defined for optimization within the bounding box:

1. Self-Region ($\mathcal{R}_{self}$): Pixels belonging to the ground truth mask of instance i .
2. Neighbor-Region ($\mathcal{R}_{neigh}$): Pixels belonging to other instances j ($j \neq i$) that fall

within the bounding box of i .

Background pixels are explicitly excluded from this loss computation. This design choice stems from the observation that the coefficient-prototype mechanism in YOLOv8 is adept at distinguishing foreground from background but prone to ambiguity when separating adjacent instances. An angular margin m is imposed to enforce separation:

$$L_{pos} = \sum_{(x,y) \in \mathcal{R}_{self}} \max(0, m - S_{i,x,y})^2$$

$$L_{neg} = \sum_{(x,y) \in \mathcal{R}_{neigh}} \max(0, S_{i,x,y} - (-m))^2$$

This ensures that the coefficient vector maximizes its directional alignment with the spatial prototypes of the target instance, while being driven towards negative correlations with interfering neighbors.

3.4.2 Adaptive Mining Strategy

A naive averaging of this loss often leads to gradient dilution due to the vast number of "easy" pixels and the spatial imbalance between self and neighbor regions. To address this, a Balanced OHEM-inspired aggregation strategy [36] is implemented. First, the mean losses for the positive (self) and negative (neighbor) terms are calculated independently. Second, explicitly following the hard mining principle to prevent gradients from being dominated by well-classified samples, a lower-bounded normalization strategy is applied. The denominator is dynamically calculated for each region as:

$$N_{denom}^k = \max(N_{hard}^k, N_{total}^k \cdot \tau), \quad \text{for } k \in \{+, -\}$$

where $k = +$ corresponds to the self-region and $k = -$ corresponds to the neighbor-region. N_{hard} represents the count of pixels violating the margin ($\text{loss} > 0$), N_{total} is the total pixel count in the respective region, and τ is a minimum keep ratio, empirically set to $\tau = 0.1$. This lower-bound constraint prevents numerical instability when hard samples are sparse while maintaining focus on boundary-critical regions. The aggregated directional objective is defined as the sum of the balanced terms:

$$\mathcal{L}_{DCL} = \frac{L_{pos}}{N_{denom}^+} + \frac{L_{neg}}{N_{denom}^-}$$

where N_{denom}^+ and N_{denom}^- represent the specific denominators calculated for the self-region and neighbor-region, respectively. Crucially, the normalization denominators are computed independently for the self-region and neighbor-region. This design ensures that the gradient contributions from both regions are balanced, preventing the optimization from being dominated by whichever region possesses a larger pixel count. Consequently, the model maintains equal sensitivity to both self-alignment and neighbor-suppression. Finally, the total objective function for the proposed framework is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{yolov8} + \lambda_{DCL} \mathcal{L}_{DCL} \quad (3.2)$$

where $\mathcal{L}_{yolov8}$ denotes the standard multi-task loss of the YOLOv8-seg baseline, comprising the box, classification, distribution focal, and mask segmentation losses, while λ_{DCL} serves as the balancing coefficient for the directional consistency loss.



Chapter 4 Experiment

4.1 Datasets

To rigorously evaluate the performance and generalization capability of the proposed cell instance segmentation model, this study employs two distinct and challenging datasets: the 2018 Data Science Bowl (DSB2018) [5] and the PanNuke dataset [6]. These datasets were selected to assess the model's robustness across different domains—specifically, general fluorescence microscopy with high biological variance (DSB2018) and complex histopathology images reflecting the "clinical wild" (PanNuke). Some examples of DSB2018 and PanNuke are presented in Figure 4.1.

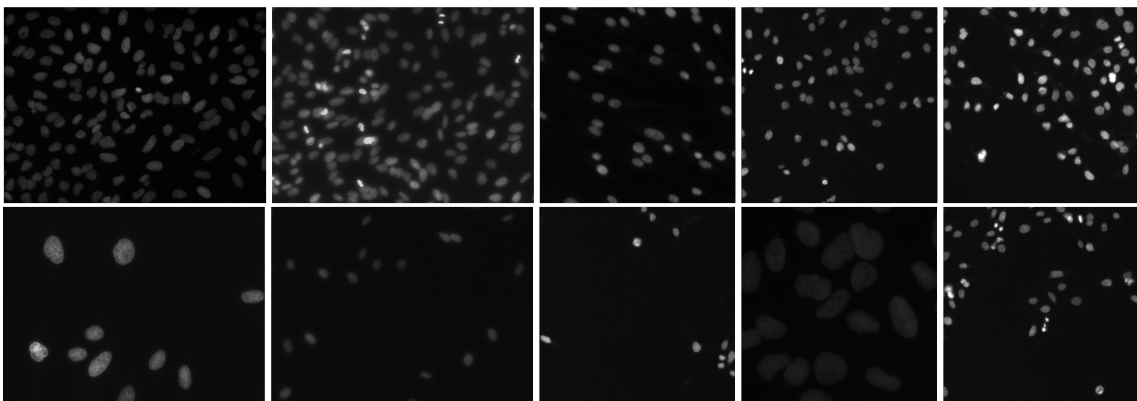
4.1.1 Data Science Bowl 2018 (DSB2018)

The Data Science Bowl 2018 dataset serves as a benchmark for evaluating an algorithm's ability to generalize across diverse biological variations. Originally released for a Kaggle challenge, the dataset is explicitly designed to prevent models from overfitting to a specific cell type or imaging modality.

The cell nuclei in this dataset are derived from a broad range of organisms, including humans, mice, and flies. Furthermore, the images capture various biological contexts



DSB2018



PanNuke

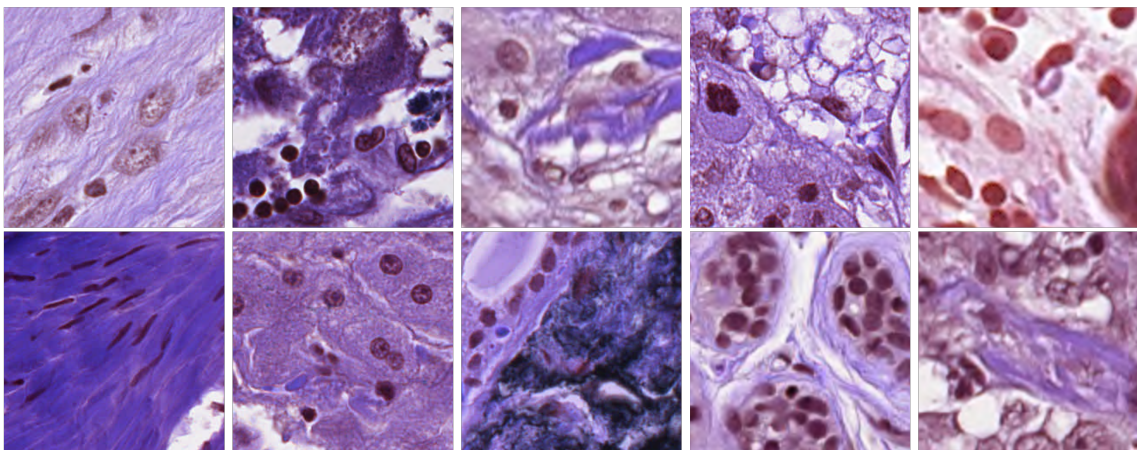
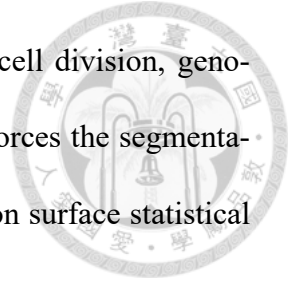


Figure 4.1: Examples of DSB2018 and PanNuke.

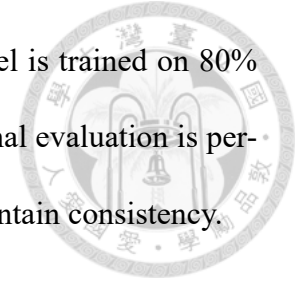
and cellular states, such as cultured monolayers, tissues, embryos, cell division, genotoxic stress, and differentiation. This high degree of heterogeneity forces the segmentation model to learn generalized cellular features rather than relying on surface statistical regularities of a single tissue type.



Data Selection and Splitting Protocol. Although the original scale of the dataset included 670 manually annotated images, this study adheres to the established data protocol used by benchmark works such as StarDist [14] and FCIS [15] to ensure fair and comparable evaluation. The protocol involves the following specifications:

- **Image Filtering:** A subset of 497 Fluorescence Microscopy images is utilized. This subset excludes images identified in previous studies [14] as containing labeling errors.
- **Resolution:** Image resolutions vary significantly, ranging from 256×256 to 520×696 pixels.
- **Dataset Split and Evaluation Strategies:** To provide both a direct benchmark comparison and a statistically robust assessment, two distinct strategies are employed for data partitioning:
 1. **Standard Benchmark Split:** Following the strict precedent set by prior works [15], the dataset is partitioned into 380 images for training, 67 for validation, and 50 for testing. This setting is used to strictly compare the results with reported baselines.
 2. **Five-Fold Cross-Validation:** To mitigate the selection bias inherent in a single validation split and verify model stability, a five-fold cross-validation strategy is implemented. The original training and validation sets are combined

into a single pool of 447 images. In each fold, the model is trained on 80% of this pool and validated on the remaining 20%. The final evaluation is performed on the same fixed set of 50 testing images to maintain consistency.



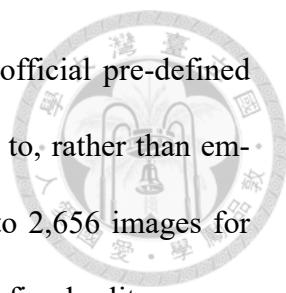
4.1.2 PanNuke Dataset

For histopathological evaluation, the PanNuke dataset is utilized, which stands as one of the largest and most diverse pan-cancer datasets for cell instance segmentation. Unlike traditional datasets that are often limited to specific organs or controlled settings, PanNuke aims to represent the "Clinical Wild". It achieves minimal selection bias by randomly sampling visual fields from over 20,000 Whole Slide Images (WSIs), thereby ensuring the inclusion of common artifacts found in routine clinical practice.

The dataset is characterized by its massive scale and diversity, containing nearly 200,000 annotated cells. These cells span 19 different tissue types (including breast, colon, lung, and kidney, among others) and are categorized into five clinically significant classes: Neoplastic, Non-neoplastic Epithelial, Inflammatory, Connective, and Dead cells. The ground truth annotations were generated through a semi-automated pipeline verified by expert clinical pathologists, ensuring high-quality labeling.

Data Selection and Splitting Protocol. The experimental setup utilizes the standard split provided by the dataset curators [6] to maintain consistency with existing benchmarks. The specifications are as follows:

- **Volume:** The dataset comprises a total of 7,901 images.
- **Resolution:** All image patches are standardized to a resolution of 256×256 pixels.

- 
- **Dataset Split:** To facilitate standardized benchmarking, the official pre-defined partitions provided by the dataset curators are strictly adhered to, rather than employing cross-validation. The dataset is explicitly divided into 2,656 images for training, 2,523 for validation, and 2,722 for testing. Using these fixed splits ensures that the results are directly comparable to existing methods.

4.2 Implementation Details

The models were developed and implemented using the PyTorch v2.5.1 deep learning framework, leveraging the CUDA 11.5 environment for accelerated computation. Training was exclusively performed on a NVIDIA RTX A6000 GPU. The utilized computing infrastructure featured an Intel Xeon Silver 4214R CPU (2.40 GHz, 48 cores), operating under the x86_64 architecture and running the Ubuntu 22.04.5 LTS operating system.

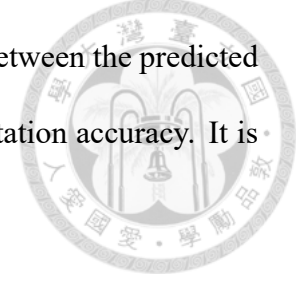
The models were trained for 50 epochs using a batch size of 6. The AdamW optimizer was employed with an initial learning rate set to 7×10^{-4} . A linear learning rate scheduler was utilized, preceded by a warmup phase of 5 epochs to stabilize training at the initial stages. The specific loss function used for optimization is shown in Eq. (3.2).

4.3 Evaluation Metrics

To strictly evaluate the proposed method, three metrics are employed: Dice Similarity Coefficient (DICE), Aggregated Jaccard Index (AJI) [37], and Panoptic Quality (PQ) [38].

Dice Similarity Coefficient (DICE)

The Dice Similarity Coefficient measures the pixel-wise similarity between the predicted mask and the ground truth, serving as a baseline for overall segmentation accuracy. It is defined as:



$$\text{DICE} = \frac{2 \times |X \cap Y|}{|X| + |Y|} \quad (4.1)$$

where X and Y denote the set of pixels in the ground truth and the predicted mask, respectively, and $|X|$ represents the cardinality (number of pixels) of the set. While effective for determining foreground coverage, DICE treats all pixels equally and does not explicitly account for individual instance separation.

Aggregated Jaccard Index (AJI)

The Aggregated Jaccard Index (AJI) is utilized as an integrated metric that simultaneously evaluates both object detection accuracy and segmentation precision. Unlike simple pixel-wise metrics, AJI aggregates the intersection metrics of matched instances while explicitly penalizing false positives and false negatives in the denominator, making it highly robust against instance-level errors such as merging or splitting. It is mathematically formulated as:

$$\text{AJI} = \frac{\sum_{i=1}^N |G_i \cap P_{j^*(i)}|}{\sum_{i=1}^N |G_i \cup P_{j^*(i)}| + \sum_{k \in U} |P_k|} \quad (4.2)$$

where G_i is the i -th ground truth object, $P_{j^*(i)}$ is the matched predicted object, and U represents the set of unmatched predicted objects (false positives).

Panoptic Quality (PQ)

Panoptic Quality offers a unified evaluation by decomposing performance into Segmenta-

tion Quality (SQ) and Recognition Quality (RQ). While RQ assesses the model's detection capability (F1-score), SQ measures the pixel-wise alignment of the successfully matched instances. It is calculated as the product of these two terms:

$$PQ = SQ \times RQ = \underbrace{\frac{\sum_{(p,g) \in TP} \text{IoU}(p, g)}{|TP|}}_{\text{Segmentation Quality (SQ)}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{Recognition Quality (RQ)}} \quad (4.3)$$

4.4 Experimental Results

This section presents a comprehensive evaluation of the proposed method on the DSB2018 and PanNuke datasets. A quantitative analysis is first conducted in Section 4.4.1 to compare performance metrics against existing methods. Subsequently, in Section 4.4.2, qualitative visual comparisons are provided to intuitively demonstrate the model's capabilities in instance separation and boundary refinement.

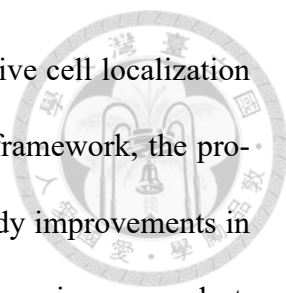
4.4.1 Quantitative Evaluation

The quantitative comparison results on the DSB2018 and PanNuke datasets are summarized in Table 4.1 and Table 4.2, respectively. The analysis focuses on the trade-offs between pixel-level accuracy and instance-level separation, as well as the adaptability of the model across different biological contexts. For the comparative analysis presented in Table 4.1 and Table 4.2, the performance metrics for the non-YOLO baseline methods—specifically HoverNet [12], DoNet [17], and FCIS [15]—are directly cited from the FCIS study [15], as they adhere to compatible evaluation protocols. In contrast, all YOLO-based models, including the baseline YOLOv8 variants and ASF-YOLO [31], were trained and evaluated in this study. Specifically regarding ASF-YOLO, although the original publica-

tion employed the DSB2018 dataset, it utilized the full dataset configuration (670 images) and primarily reported mean Average Precision (mAP). To guarantee a fair assessment under identical experimental constraints, ASF-YOLO was retrained using its official implementation on the standardized subset (497 images) used in this work, allowing for valid comparison of segmentation metrics (DICE, AJI, and PQ).

Instance Separation in Dense Scenarios (DSB2018). As presented in Table 4.1, the DSB2018 dataset serves as a rigorous benchmark for evaluating segmentation performance in scenarios with densely clustered cells. A nuanced comparison between the baseline YOLOv8l-seg and the FCIS [15] reveals distinct performance characteristics. It is observed that while the baseline achieves a competitive PQ (0.775) slightly surpassing FCIS (0.770), its AJI score (0.803) significantly trails behind FCIS (0.828). This discrepancy is critical: PQ averages the quality of matched instances and can be buoyed by high performance on isolated cells, whereas AJI heavily penalizes pixel-level mismatches caused by spillover errors. The lower AJI of the baseline confirms its tendency to exhibit mask spillover between touching instances. In contrast, the proposed method effectively addresses this limitation, achieving the highest AJI (0.832) and PQ (0.816). Although FCIS attains a marginally higher DICE score (0.939) compared to the proposed method (0.929), the proposed approach surpasses FCIS in AJI. This indicates that while FCIS excels at general foreground pixel coverage, the proposed method demonstrates superior capability in instance separation, successfully resolving the boundaries of touching cells where mask spillover typically occurs in baseline models.

Detection Robustness in Complex Scenes (PanNuke). Turning to the PanNuke dataset, a general trend is observed favoring detection-oriented architectures. YOLO-based approaches demonstrate favorable performance compared to non-YOLO methods,



as the inherent robustness of the YOLO framework allows for effective cell localization and background suppression in clinical scenarios. Building on this framework, the proposed method achieves the highest PQ score through subtle and steady improvements in both recognition and segmentation quality. Although the performance gains are modest, the consistent results across metrics indicate a reliable capability in identifying instances and delineating their masks. This suggests that the proposed modifications enhance the overall robustness of the model when facing the varied conditions of the "clinical wild".

Comparative Analysis on Improvement Margins and Efficiency. To further validate the architectural contributions, the proposed method is compared against the cell-task-oriented ASF-YOLO [31] and the large-scale YOLOv8x-seg.

First, regarding ASF-YOLO, which integrates specialized attention mechanisms (CPAM) and multi-scale fusion explicitly for cell segmentation, a notable limitation is observed. As shown in Tables 4.1 and 4.2, ASF-YOLO fails to outperform the general-purpose baseline on both datasets. This underperformance suggests that merely enhancing feature extraction via attention mechanisms is insufficient to resolve the specific challenge of "touching borders" in cytology. While attention helps the model focus on cellular features, it lacks an explicit mechanism to enforce boundary separation. In contrast, the proposed method outperforms ASF-YOLO by a significant margin. This validates that the proposed design—specifically the HRPN for spatial granularity and the DCL for directional repulsion—addresses the physical bottlenecks of instance separation more effectively than generic feature refinement.

Second, the proposed method is compared against YOLOv8x-seg. As the largest variant in the YOLOv8 family, YOLOv8x utilizes a significantly deeper and wider back-

bone with 71.75M parameters. However, despite this massive increase in model capacity, it does not yield better segmentation performance than the proposed method (56.74M parameters), which is constructed upon the lighter YOLOv8l backbone. This indicates that simply scaling up the model size reaches a point of diminishing returns and cannot substitute for the targeted architectural designs required for fine-grained cell delineation.

Table 4.1: **Quantitative comparison on DSB2018.** The table evaluates methods under two settings: Standard Benchmark Split and 5-Fold Cross Validation. The best results are highlighted in **bold**, and the second-best results are underlined.

Method	#Params (M)	DICE $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
<i>Setting 1: Standard Benchmark Split</i>				
HoverNet [12]	49.70	0.898	0.762	0.762
DoNet [17]	67.71	0.823	0.716	0.673
FCIS [15]	39.75	0.939	<u>0.828</u>	0.770
ASF-YOLO [31]	48.47	0.904	0.782	0.742
YOLOv8l-seg (Baseline)	45.94	0.914	0.803	0.775
YOLOv8x-seg	71.75	0.913	0.807	<u>0.779</u>
Proposed Method	56.74	<u>0.929</u>	0.832	0.816
<i>Setting 2: 5-Fold Cross Validation</i>				
ASF-YOLO	48.47	0.902	0.780	0.748
YOLOv8l-seg (Baseline)	45.94	0.911	0.802	0.777
YOLOv8x-seg	71.75	0.912	0.804	0.778
Proposed Method	56.74	0.927	0.829	0.814

Table 4.2: **Quantitative comparison on PanNuke.** The best results are highlighted in **bold**, and the second-best results are underlined.

Method	DICE $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
HoverNet [12]	0.798	0.646	0.595
DoNet [17]	0.781	0.612	0.544
FCIS [15]	0.816	0.653	0.610
ASF-YOLO [31]	<u>0.826</u>	0.670	0.657
YOLOv8l-seg	0.823	0.673	0.662
YOLOv8x-seg	0.825	<u>0.675</u>	<u>0.667</u>
Proposed Method	0.828	0.685	0.687

4.4.2 Qualitative Evaluation



To intuitively demonstrate the superiority of the proposed method, qualitative comparisons are provided across three figures: Figure 4.2 and Figure 4.3 for the DSB2018 dataset, and Figure 4.4 for the PanNuke dataset. Across all figures, the columns from left to right correspond to the Input Image, Ground Truth, predictions from the baseline (YOLOv8l-seg), the YOLO-based competitor (ASF-YOLO), and the proposed method.

Visual Analysis on DSB2018. Qualitative comparisons are presented in Figure 4.2 and Figure 4.3. Figure 4.2 provides a detailed examination, where the first row displays the full field of view and the second row offers a magnified view of the region enclosed in the red box. As visualized in the zoomed-in regions, a significant disparity in segmentation quality is observed between the proposed method and other YOLO-based models. In crowded regions (highlighted by red arrows), the baseline YOLOv8l-seg and ASF-YOLO frequently exhibit boundary artifacts. Specifically, two types of errors are prevalent: (1) "mask spillover," where the mask of one instance erroneously expands into an adjacent instance due to bounding box overlap, and (2) imprecise boundary delineation, where the predicted contours fail to tightly adhere to the cellular shape. These visual artifacts corroborate the quantitative analysis regarding the limitations of standard YOLO architectures in dense scenarios. In contrast, the proposed method successfully delineates clear and precise boundaries between touching instances. It effectively mitigates the spillover effect and refines boundary adherence, yielding segmentation results that closely align with the Ground Truth. Finally, Figure 4.3 presents additional examples from the test set, further confirming the consistency of these improvements across diverse dense scenarios.

Visual Analysis on PanNuke. The results on the PanNuke dataset, presented in

Figure 4.4, highlight the model’s capability to handle the complexities of the ”clinical wild”. As indicated by the red arrows, YOLOv8l-seg and ASF-YOLO exhibit detection errors, such as false negatives and false positives. These inconsistencies stem from the loss of fine-grained cues in standard downsampling. Although the HRPN is primarily designed for mask generation, its integration enforces the shared backbone to learn more robust, high-resolution representations during joint optimization. This enriched feature extraction empowers the detection head to better distinguish cells from background noise. Consequently, the proposed method mitigates these detection errors while maintaining superior mask precision compared to other models, contributing to the steady improvements observed in PQ metrics.

In summary, the qualitative visualizations are consistent with the quantitative metrics reported in Section 4.4.1. The proposed method demonstrates a robust capability to separate touching cells in dense scenarios while preserving morphological details in complex backgrounds, validating its effectiveness across diverse biological contexts.

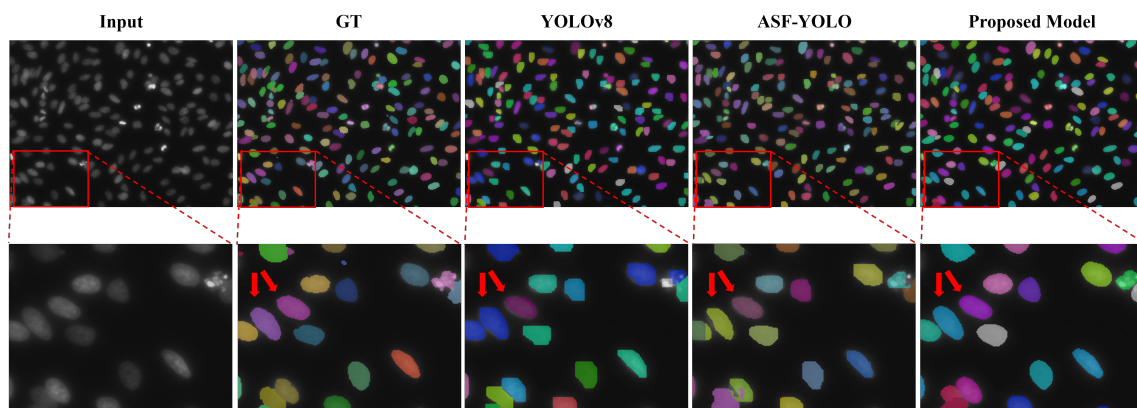


Figure 4.2: **Qualitative comparison of instance separation in dense clusters (DSB2018).** The second row presents the magnified view of the region outlined in the red box. Red arrows highlight significant segmentation artifacts (merging and spillover) in the baseline and ASF-YOLO models, which are effectively resolved by the proposed method.

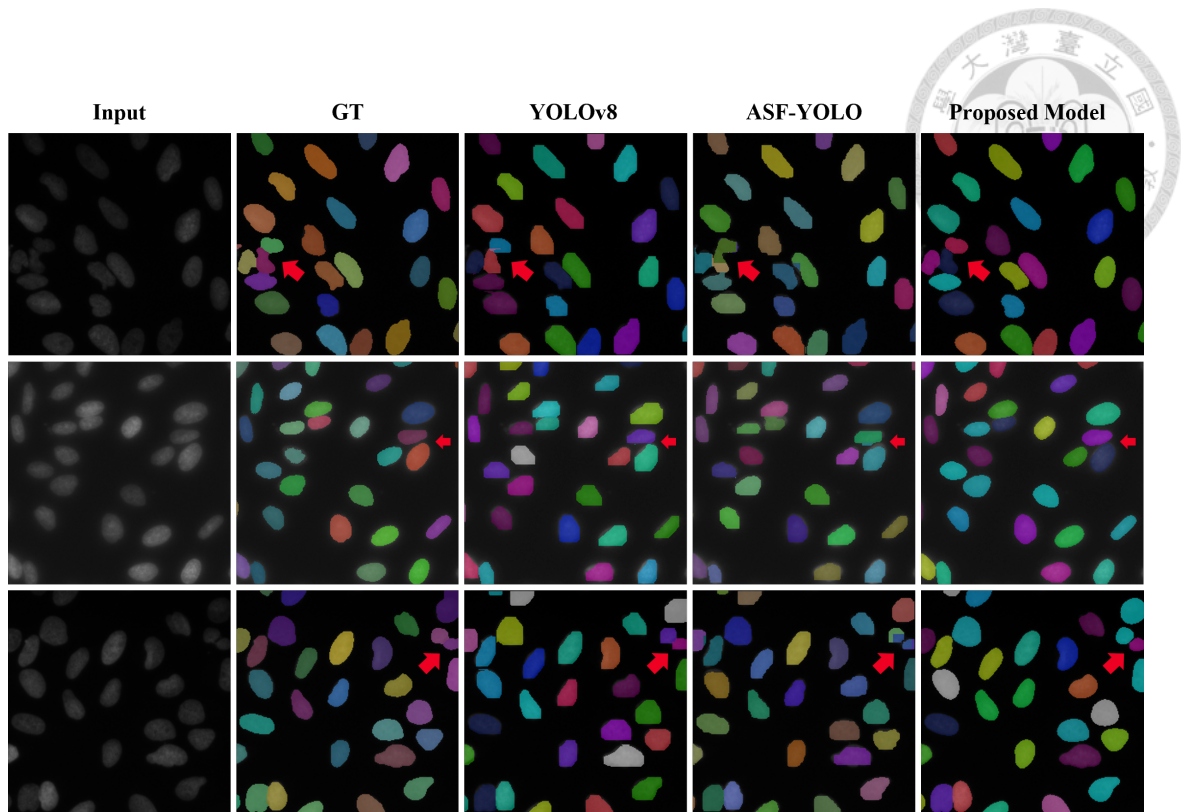


Figure 4.3: **Supplementary qualitative comparisons on the DSB2018 dataset.** Additional examples demonstrating the consistent robustness of the proposed method in delineating boundaries and separating instances across diverse dense scenarios.

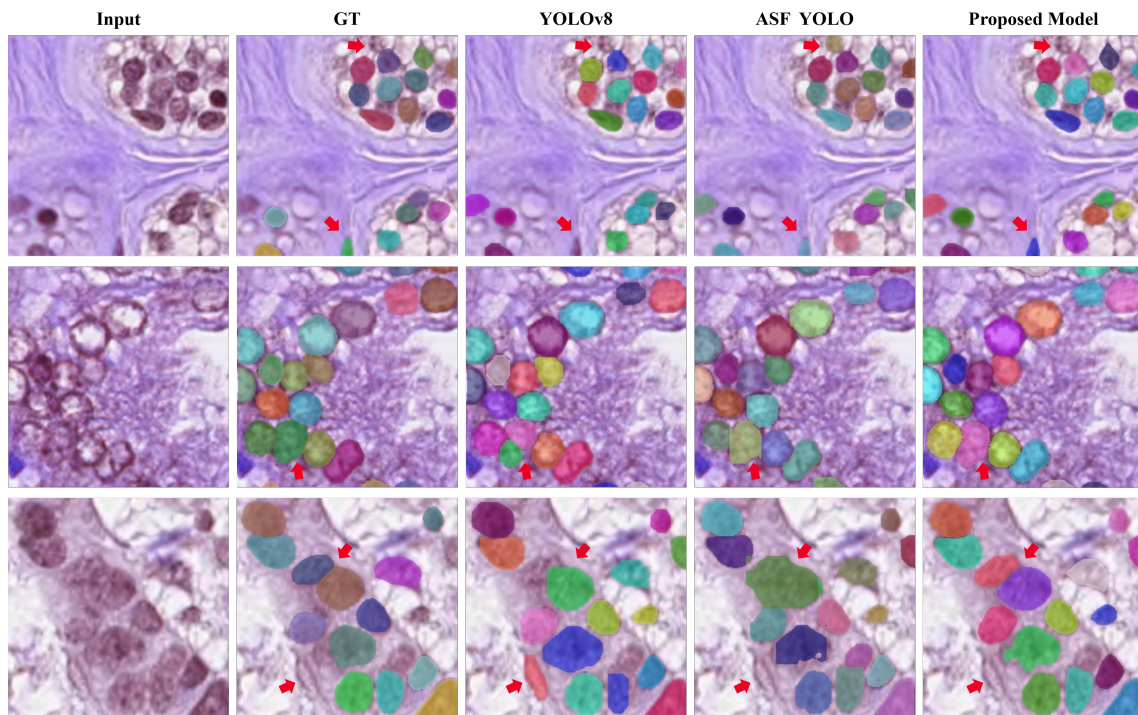


Figure 4.4: **Qualitative comparison on the PanNuke dataset.** From left to right: Input Image, Ground Truth, YOLOv8l-seg, ASF-YOLO, and Proposed Method.



4.5 Ablation Study

To verify the effectiveness of the proposed modules and justify the architectural design choices, a series of ablation experiments were conducted. Unless otherwise specified, all experiments in this section were performed on the DSB2018 dataset using the Standard Benchmark Split (380 training, 67 validation, 50 testing images) to ensure consistent and comparable evaluation conditions.

4.5.1 Effectiveness of Proposed Components

The impact of progressively integrating the three core contributions into the baseline YOLOv8l-seg model was investigated. For clarity in the following discussions and tables, the proposed components are denoted as follows:

- HRPN: High-Resolution Prototype Network, replacing the standard prototype branch.
- FiLM: Proto-Guided Coefficient Generation module.
- DCL: Directional Consistency Loss, the auxiliary directional loss function.

The quantitative results of the stepwise integration are summarized in Table 4.3.

High-Resolution Prototype Network. The introduction of HRPN brings comprehensive improvements across all metrics. Notably, the Panoptic Quality (PQ) yields the most substantial gain, rising from 0.775 to 0.808 (+3.3%), while AJI and DICE improve to 0.825 and 0.927, respectively. This indicates that the standard FPN-based prototype branch sacrifices too much spatial information. By preserving fine-grained spatial details, the HRPN not only refines mask boundaries but also enhances the overall reliability of instance recog-

dition and segmentation.

Proto-Guided Coefficient Generation (FiLM). Integrating the FiLM mechanism further improves the PQ to 0.812. Although this module does not directly alter the bounding box predictions, it enhances the structural consistency between the generated coefficients and the prototypes. By conditioning the coefficient head on the spatial priors of the prototypes, the model ensures that the generated masks are more compatible with the detected objects. This improved alignment helps convert marginal segmentation predictions into valid true positives, thereby contributing to the increase in the PQ score.

Directional Consistency Loss (DCL). Finally, the addition of the DCL boosts the AJI to 0.832 and PQ to 0.816. The improvement in AJI is particularly critical as this metric strictly penalizes boundary spillover between adjacent instances. Unlike standard segmentation objectives that primarily focus on pixel-wise classification accuracy without explicitly regulating the angular relationship between instance embeddings, the proposed DCL explicitly pushes the feature vectors of touching neighbors towards negative cosine similarity. This strong repulsive force effectively resolves the "touching cell" problem, ensuring clear separation even in dense clusters.

Table 4.3: **Ablation study of component effectiveness.** Comparison of the baseline with progressively added modules on the DSB2018 dataset. Best results are **bolded**.

Method	DICE $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
Baseline (YOLOv8l-seg)	0.914	0.803	0.775
+ HRPN	0.927	0.825	0.808
+ HRPN + FiLM	0.926	0.825	0.812
+ HRPN + FiLM + DCL	0.929	0.832	0.816

4.5.2 Impact of Interaction Scale in SAF Block

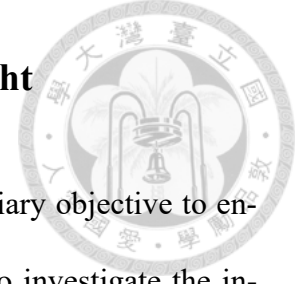


In the Split-Attention Fusion (SAF) block, the interaction branch aggregates context by downsampling features to a unified resolution before concatenation and gating. To determine the optimal granularity for this global context modeling, three different interaction scales were evaluated: High (Stride 4), Medium (Stride 8), and Low (Stride 16). The results are presented in Table 4.4. Experimental results indicate that aggregating context at the lowest resolution yields the best performance across all metrics. Conversely, utilizing high-resolution features for the interaction branch results in a performance drop. This finding can be attributed to the nature of the Split-Attention mechanism described in Section 3.2. The primary goal of the interaction branch is to capture global semantic dependencies to modulate the local details preserved in the retention branch. A lower resolution effectively expands the receptive field, allowing the Dilated Context Gating module to integrate information from a broader spatial context. In contrast, maintaining the interaction at an excessively high resolution limits the effective receptive field to local neighborhoods. This prevents the model from capturing the broader contextual information—such as tissue texture and cluster distribution—which is essential for distinguishing genuine cells from background artifacts and resolving ambiguities in dense regions. Thus, a Stride of 16 provides the optimal balance, ensuring the gating weights reflect global structural priors.

Table 4.4: **Impact of interaction resolution in SAF Block.** Aggregating global context at the lowest resolution (Low) proves most effective.

Interaction Scale	DICE $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
High (Stride 4)	0.924	0.824	0.808
Medium (Stride 8)	0.928	0.826	0.810
Low (Stride 16)	0.929	0.832	0.816

4.5.3 Sensitivity Analysis of the Auxiliary Loss Weight



The Directional Consistency Loss (DCL) functions as an auxiliary objective to enforce directional alignment between coefficients and prototypes. To investigate the influence of this constraint on the overall optimization, the sensitivity of the model to the balancing weight λ_{DCL} was analyzed. The case where $\lambda_{DCL} = 0$ corresponds to the model configuration without the auxiliary loss (consistent with the result in Table 4.3). As shown in Table 4.5, increasing λ_{DCL} from 0 to 1 yields a steady improvement in AJI and PQ, confirming that explicitly enforcing directional consistency is crucial. However, increasing the weight beyond this point leads to a noticeable degradation in segmentation performance. This trend suggests a trade-off between the primary segmentation objective and the auxiliary directional constraint. When λ_{DCL} is excessively large, the gradients from the DCL term tend to dominate the optimization process, potentially interfering with the minimization of the primary Binary Cross-Entropy loss. Specifically, while the BCE loss is prone to the geometric ambiguity discussed in Section 3.4 (prioritizing magnitude over direction), an excessively high λ_{DCL} forces the network to over-correct by ignoring magnitude entirely. Therefore, a balanced weight of 1 is optimal to mitigate this ambiguity without compromising the confidence estimation.

Table 4.5: **Sensitivity analysis of the auxiliary loss weight (λ_{DCL}).** A balanced weight of 1 yields the best performance.

Weight (λ_{DCL})	DICE $\uparrow$	AJI $\uparrow$	PQ $\uparrow$
0	0.926	0.825	0.812
1	0.929	0.832	0.816
2	0.926	0.827	0.808
3	0.923	0.823	0.806



Chapter 5 Conclusion

This thesis has presented a comprehensive framework for high-precision cell instance segmentation, re-engineering the YOLOv8-seg architecture to address the unique challenges of biomedical microscopy. While unified, one-stage detectors offer significant advantages in computational efficiency, their direct application to this domain has historically been hampered by architectural trade-offs that prioritize semantic abstraction over spatial fidelity. The research conducted in this study bridges this gap, demonstrating that the speed of YOLO-based models can be reconciled with the rigorous accuracy requirements of computational pathology. Central to this contribution is the proposal of three targeted enhancements: the High-Resolution Prototype Network, the Proto-Guided Coefficient Generation mechanism, and the Directional Consistency Loss. By reconstructing the prototype branch to preserve fine-grained spatial details, establishing a structural condition between the detection head and mask generation, and enforcing directional alignment between prototypes and coefficients, the proposed method effectively resolves the critical issues of coarse boundaries and instance merging that plague standard architectures.

The effectiveness of these contributions is substantiated by rigorous empirical evaluation on the Data Science Bowl 2018 and PanNuke datasets. Quantitative results indicate that the proposed framework delivers superior performance compared to established base-

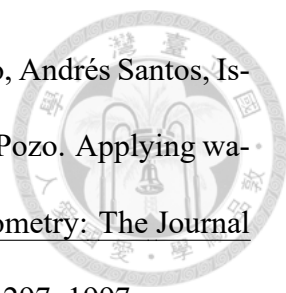
lines, most notably outperforming the standard YOLOv8-seg architecture. Specifically, in metrics that penalize instance-level errors such as the Aggregated Jaccard Index and Panoptic Quality, the model demonstrated a robust capability to separate tightly clustered cells without the need for complex post-processing. These findings confirm that the introduction of high-resolution features and the auxiliary directional loss does not merely offer incremental improvements but fundamentally enhances the discriminative power of the model in dense biological environments.

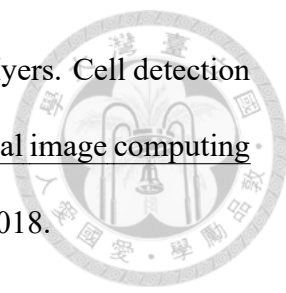
Despite the promising results presented in this thesis, the current framework relies on fully supervised training with dense pixel-level annotations. Creating such high-quality datasets is prohibitively expensive and time-consuming, particularly in computational pathology where datasets like PanNuke contain hundreds of thousands of cells. The sheer volume of instances creates a significant "annotation bottleneck." Therefore, a primary direction for future research is to transition towards semi-supervised learning paradigms. The objective is to develop a framework capable of generalizing from a small fraction of fully annotated data (e.g., 10%) to a large volume of unlabeled images. The High-Resolution Prototype Network proposed in this study offers a unique advantage for this task. Since the prototypes effectively encode global shape priors from the limited labeled set, they can serve as robust templates to guide segmentation on unlabeled samples. Future work will investigate strategies to leverage these high-fidelity prototypes for generating reliable pseudo-labels or enforcing structural consistency on unlabeled data, thereby significantly reducing the annotation burden while maintaining high segmentation precision.



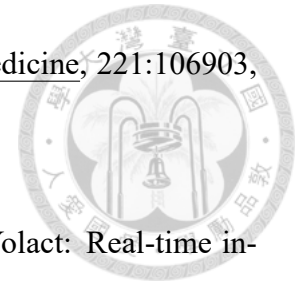
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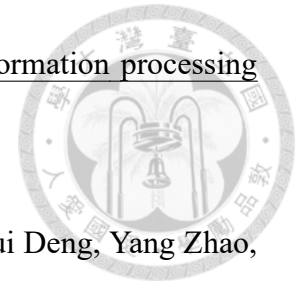
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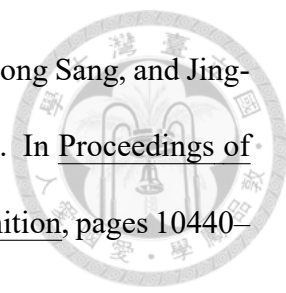


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